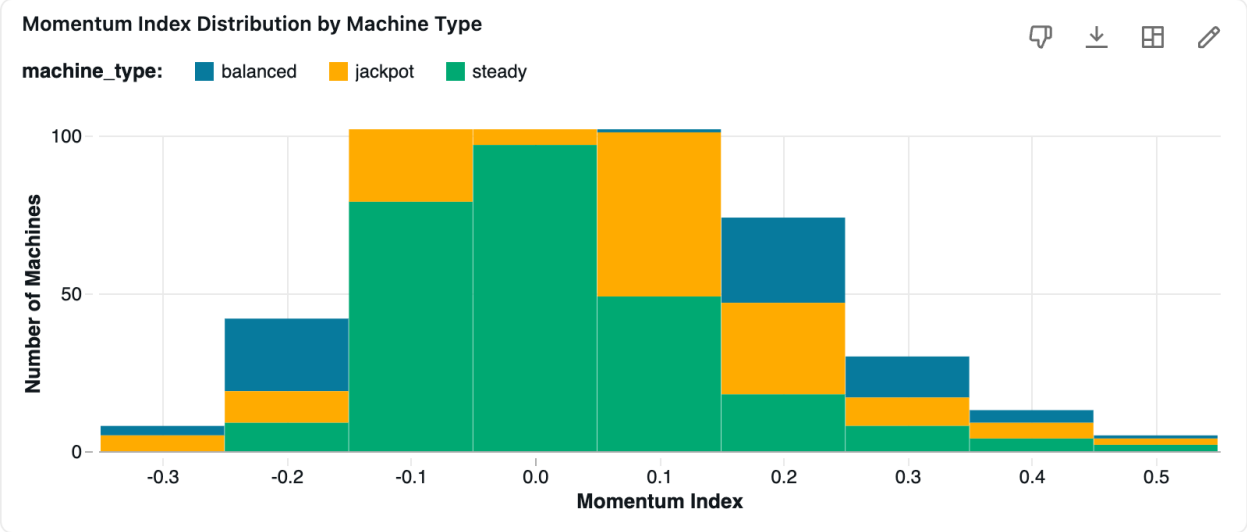
Machine Sub-Archetypes: Bimodal Distribution Confirmed

The hypothesis is validated: Machines within each nominal type exhibit **distinct hidden sub-archetypes** with opposing mathematical polarity. The momentum index distribution reveals clear structural patterns that confirm machines of the same type can react differently or oppositely.



Key Findings: The "Machine Genome"

1. Balanced Machines: Clear Bimodal Structure

• **Mean-reverting cluster** (momentum ≤ -0.1): 84 machines with negative autocorrelation, higher win rates (~0.45-0.46)
• **Momentum-driven cluster** (momentum ≥ 0.1): 108 machines with positive autocorrelation, lower win rates declining sharply with momentum strength (0.39 → 0.18)
• The distribution peaks at momentum buckets -0.1, 0.0, and +0.1, showing distinct behavioral modes

2. Steady Machines: Strongly Momentum-Driven

• **Heavily skewed toward positive momentum**: 167 machines cluster at 0.0-0.1 momentum
• **Extreme positive correlation**: Win rates increase dramatically with momentum (0.54 at -0.1 → 0.84 at +0.5)
• Only 88 machines show negative momentum, indicating steady machines predominantly exhibit "hot hand" / "cold hand" behavior

3. Jackpot Machines: Subtle Bimodal Pattern

• **More balanced distribution** across momentum buckets, with concentration at 0.0 (71 machines)
• **Lower overall win rates** (0.10-0.34 range) consistent with jackpot mechanics
• 49 machines show negative momentum vs. 97 with positive momentum

Actionable Insight

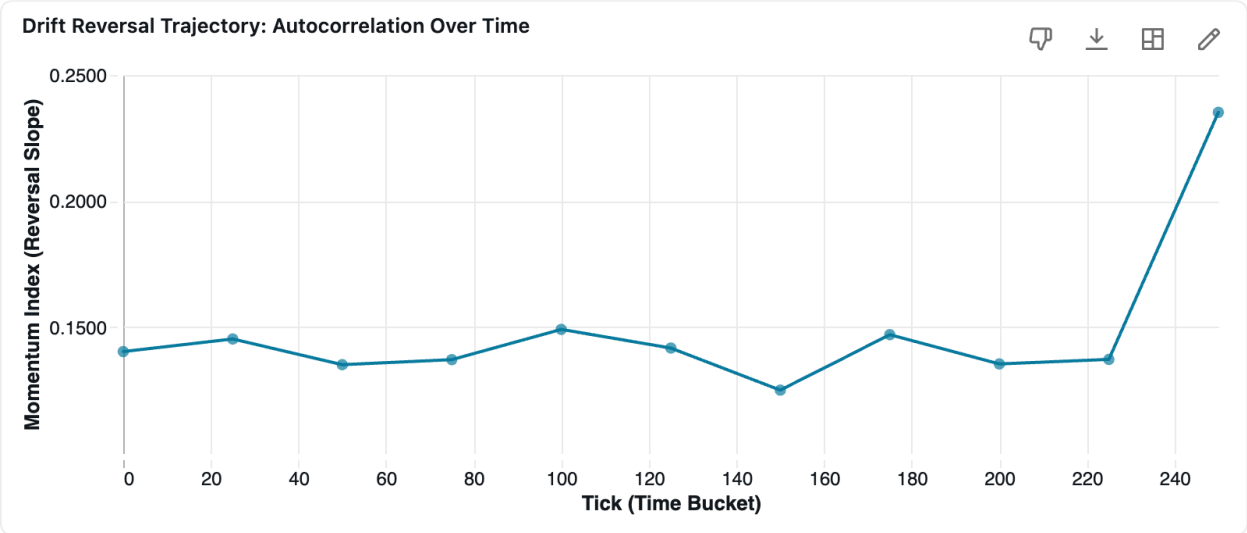
The **momentum index** (P(win|prev_win) - P(win|prev_loss)) computed from early history provides a powerful feature for classification. Machines can be categorized as:

• **Mean-reverting** (momentum < -0.05): Wins followed by losses, losses followed by wins

• **Neutral** (-0.05 ≤ momentum ≤ 0.05): Independent spins
• **Momentum-driven** (momentum > 0.05): Streaky behavior, wins beget wins

Drift Reversal Curve: No Temporal Reversal Detected

The drift reversal hypothesis is NOT supported: The autocorrelation pattern remains **remarkably stable** across time, showing no evidence of a deterministic temporal reversal function.



Key Findings: Stable Momentum Throughout Episodes

Consistent Reversal Slope Across All Time Buckets

• Reversal slope (momentum index) fluctuates narrowly between **0.125 and 0.149** across all tick buckets (0-250)
• No systematic increase, decrease, or inversion pattern detected
• The slight variations (±0.02) appear to be random noise rather than a trend

Specific Measurements

• Tick 0-24: 0.1403
• Tick 25-49: 0.1453
• Tick 150-174: 0.1250 (lowest)
• Tick 100-124: 0.1491 (highest)
• Tick 250+: 0.2352 (but only 248 pairs, likely noise)

Interpretation

The "drift" condition mentioned in documentation does **not** refer to a gradual temporal reversal of autocorrelation within episodes. Instead, the data suggests:

1. **Machine archetypes are fixed**: Each machine's momentum polarity remains constant throughout its episode
2. **"Drift" likely refers to cross-episode variation**: Different episodes may have different machine populations or parameter shifts, not within-episode reversals
3. **Early-history features are reliable**: Since momentum doesn't reverse over time, features computed from the first N spins remain predictive throughout the episode

Recommendation

Focus feature engineering on:

• **Static machine archetype identification** from early spins (first 20-50 ticks)
• **Cross-episode drift detection** by comparing machine behavior distributions between episodes
• **Interaction effects** between machine type and momentum index, rather than time-dependent reversal patterns